Basics of Computers

Ilaria Tiddi

16. September 2024

Recap

Lectures

Guest lectures (70% of the exam questions)

Still to come:

- Peter Bloem (Machine Learning)
- Guszti Eiben (Computational Intelligence)
- Koen Hindriks (Social AI)
- Stefan Schlobach (Ethical AI)
- Benno Kruit (Data processing and Engineering)
- Ilaria Tiddi (Hybrid Intelligence)

Lectures

Methodology Lectures (30% of the exam questions)

Steps

1. ~~Determining the problem~~
2. ~~Finding Literature~~
3. Basics of computers
4. Methods in AI (approach/solution)
5. Empirical Evaluations
6. Mathematics for AI (Discrete)

Final (Report) Assignment

12 pages scientific report (LNCS template)

- Abstract (1 page)
- Introduction (1 page)
- What is the problem? (2 pages) ⟵ *just submitted*
- What has been done? (3 pages) ⟵ *literature due Sept 22*
- How can the problem be solved? (3 pages max) ⟵ approach/*method due Sept 29*
- Is the solution valid? (2 pages) ⟵ *evaluation due Oct 6*
- Conclusion (1page)

Where are we?

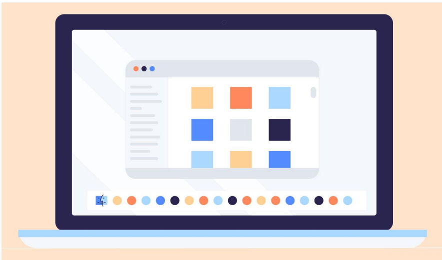
Where are we?

1. Defining the problem
2. Finding related work
3. **Building an AI system**
4. Evaluation of an AI system

What do you need to build an AI system?

What are these?

A



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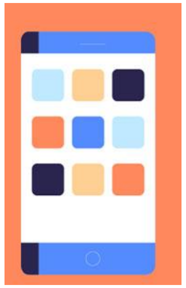

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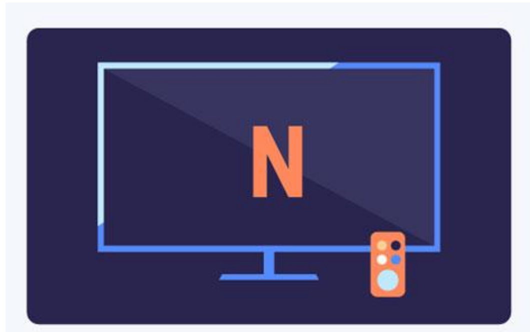

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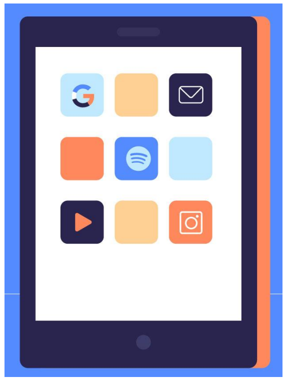

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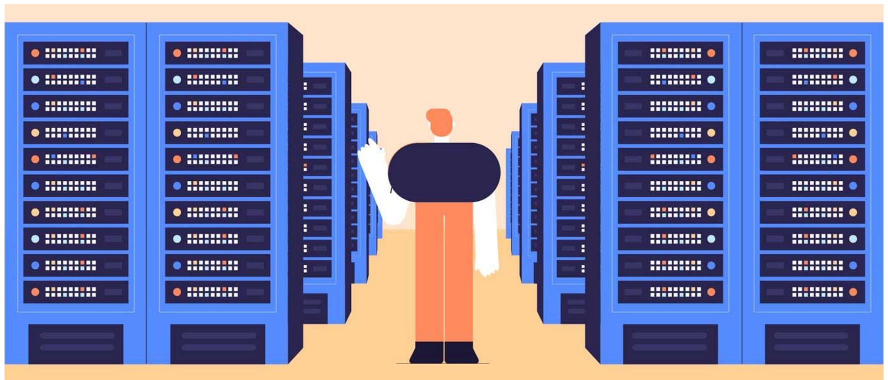
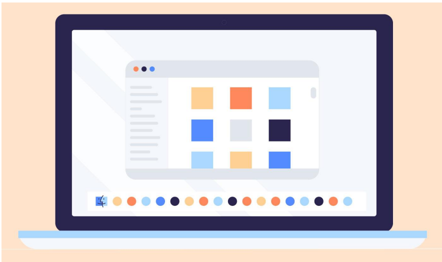


What do they have in common?

A



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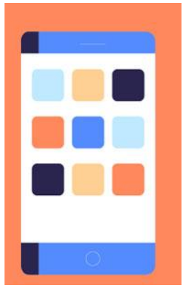

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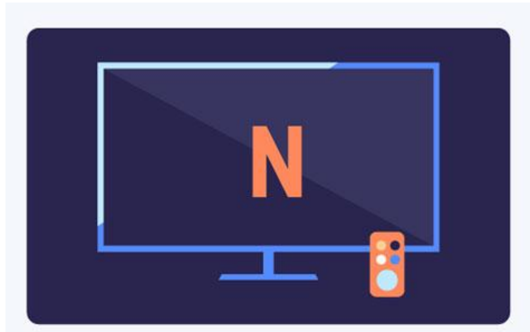

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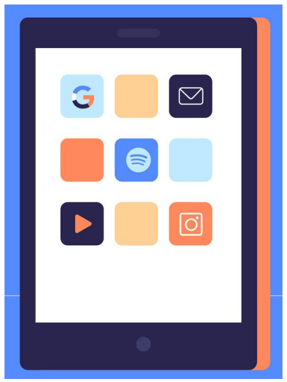

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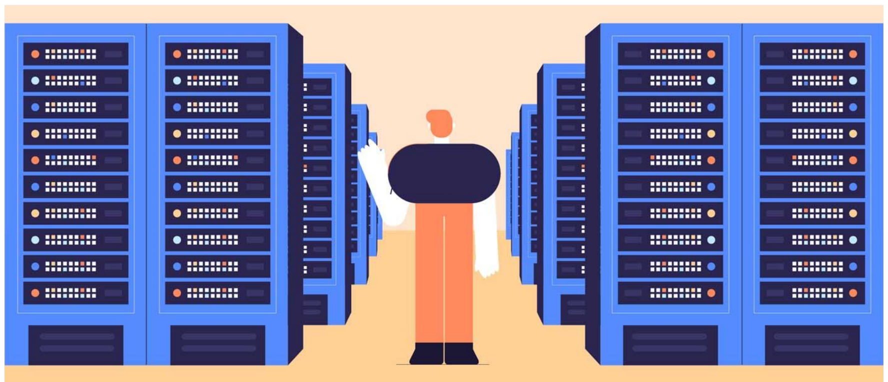
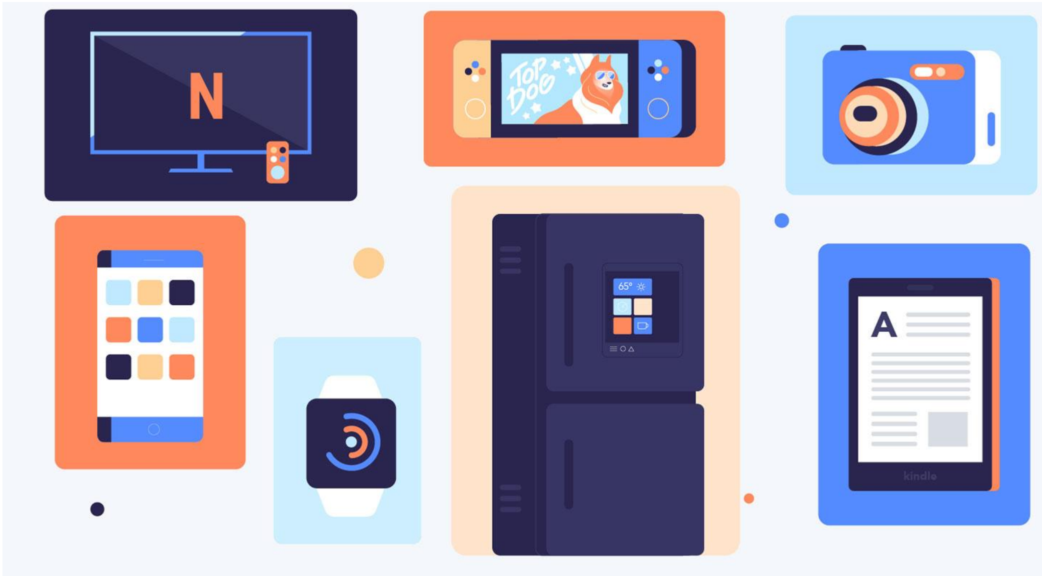


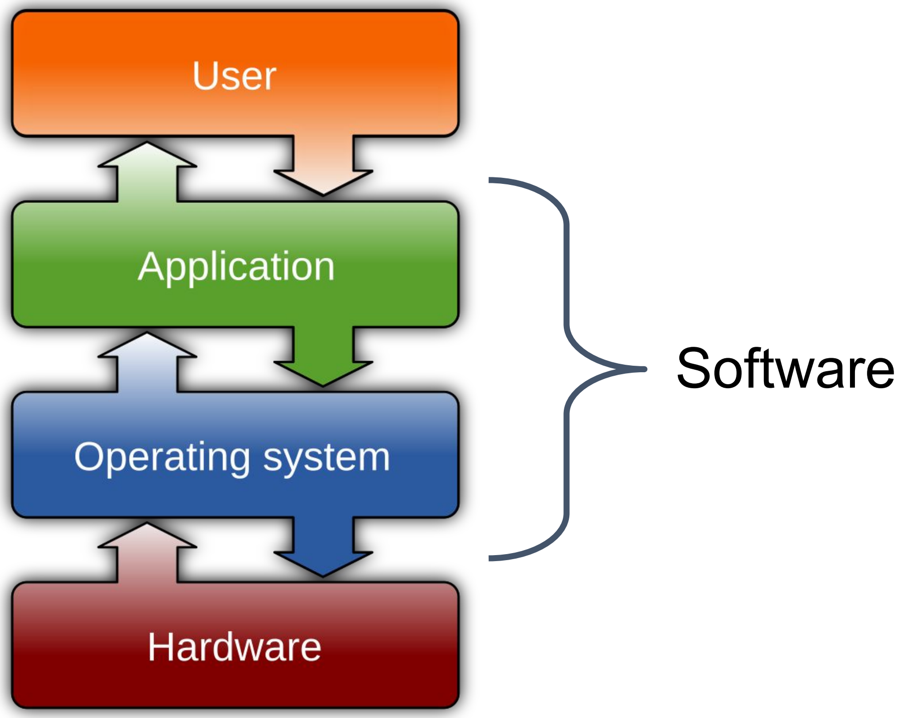
Computers

Any system device that manipulates (= stores, retrieves, processes) data.



Source : https://edu.gcfglobal.org/en/computerbasics/what-is-a-computer/1/

Computer Architecture



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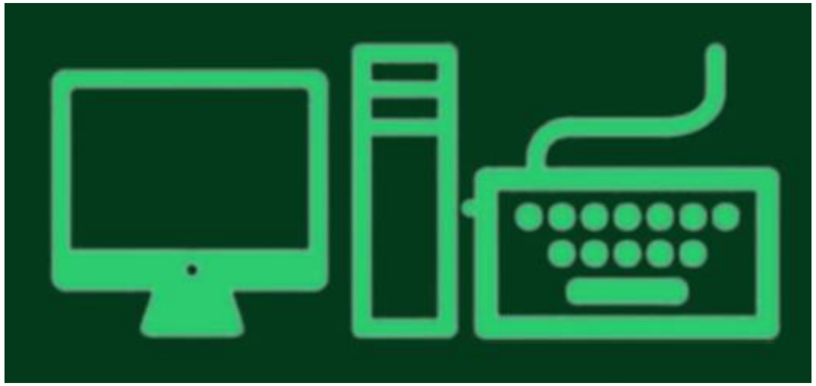
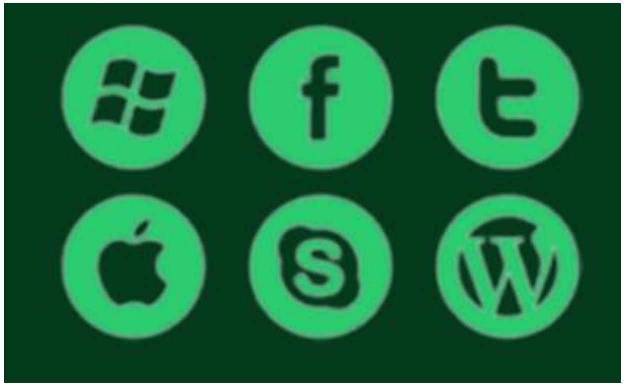
Hardware vs. Software?



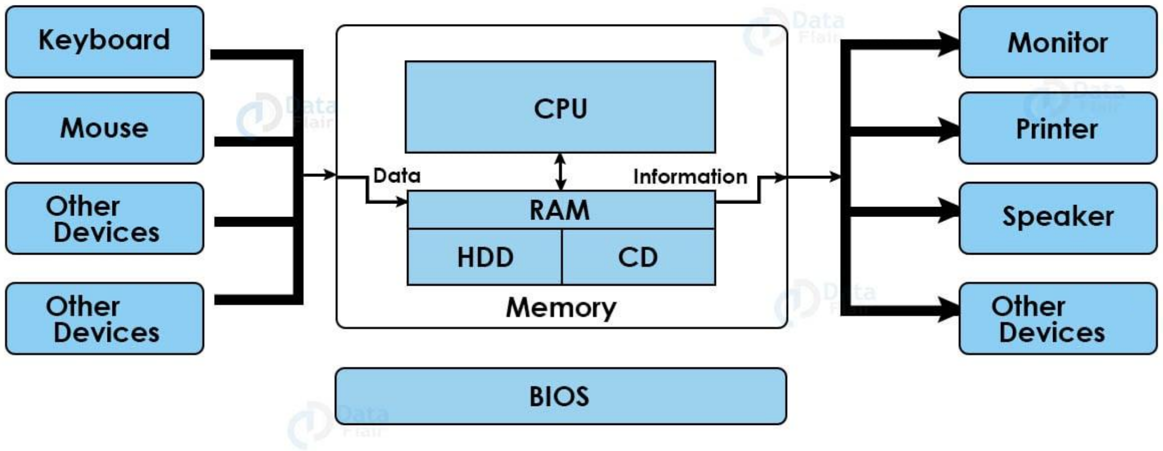


Source : https://www.lifewire.com/computer-hardware-2625895

Hardware

1. Peripheral devices
2. Input devices
3. Output devices
4. Internal

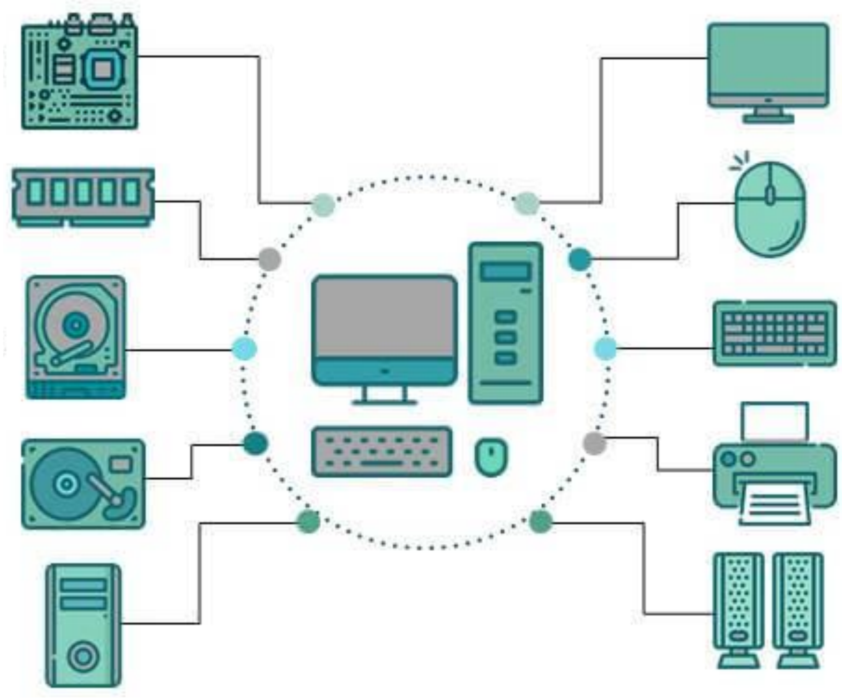
   Components



Source : https://data-flair.training/blogs/basics-of-computer-hardware-and-software/

Hardware

1
2
3
4
5



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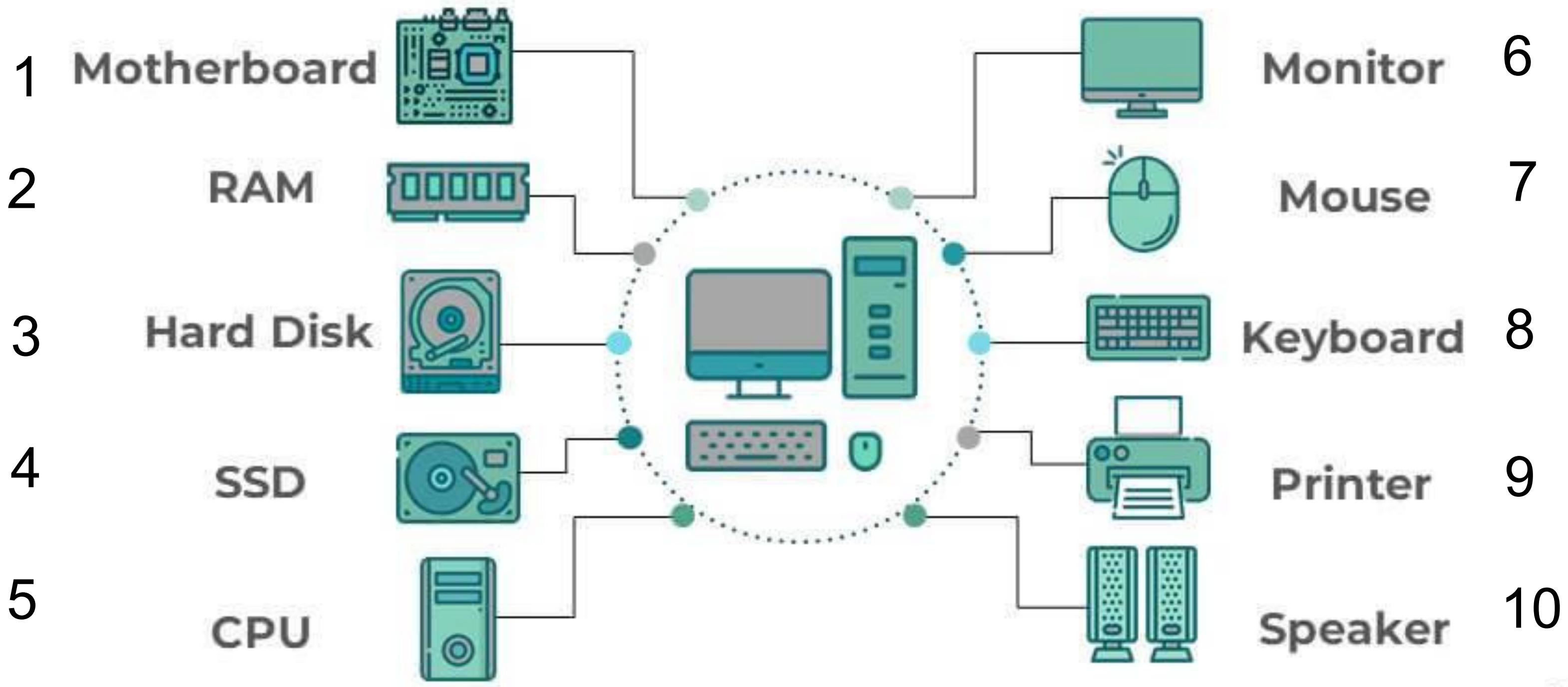
Source : https://www.educba.com/types-of-computer-hardware/

Hardware



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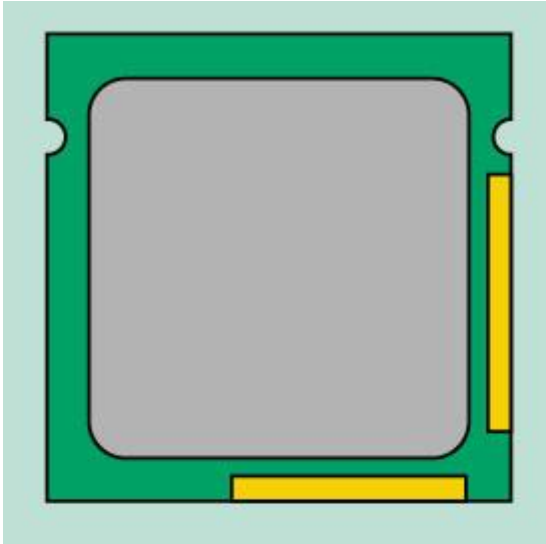
Hardware

Processors/ CPUs

- **calculations and operations** on the data
- execute instructions
- Arithmetic and Logical Unit **ALU**: (addition, subtraction, multiplication, division and also making logical decisions like > or < , etc)
- Control Unit **CU**: execution of the tasks



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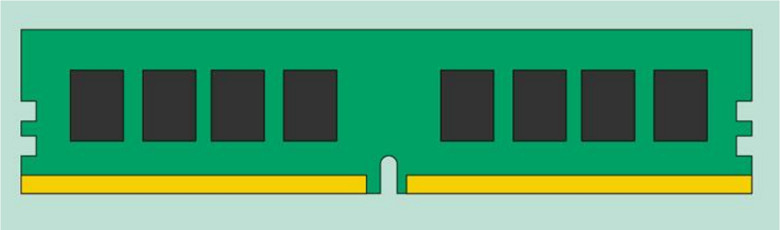

Memory / **RAM**:

- Main, primary memory of the computer
- Stores data and instructions, while the processing is taking place in the CPU.
- volatile (data is lost when the power is disconnected)
- How do you measure RAM ?

Another primary memory is ROM (Read Only Memory).

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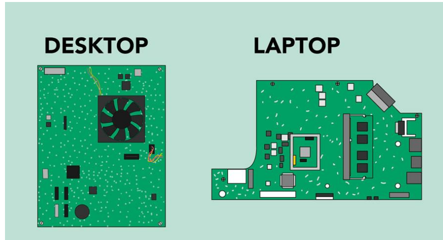
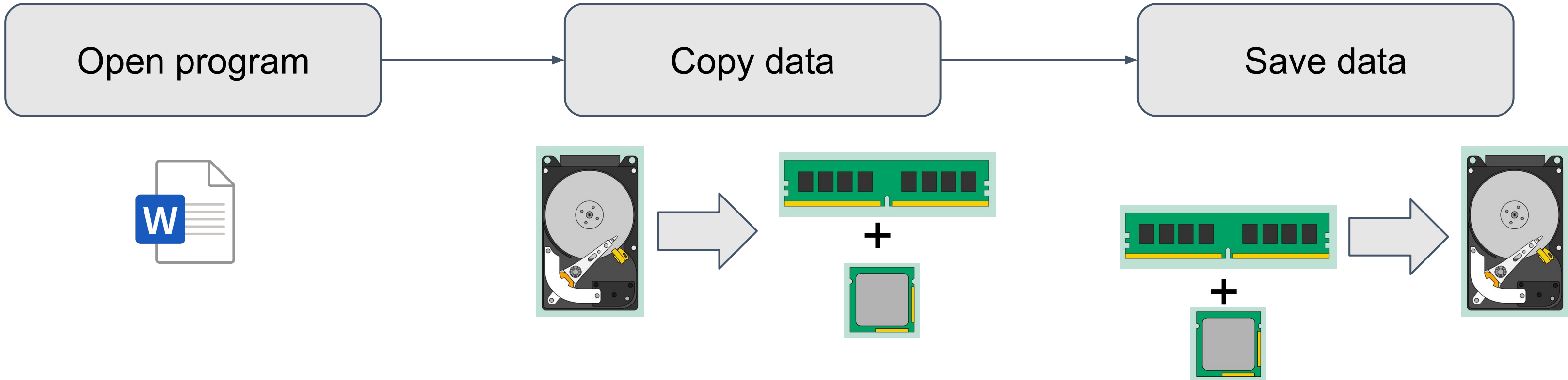
Hardware



- Motherboard: holds CPU, GPU, RAM, connectors
- Hard Disk & Solid State Disk (SSD): long state storage



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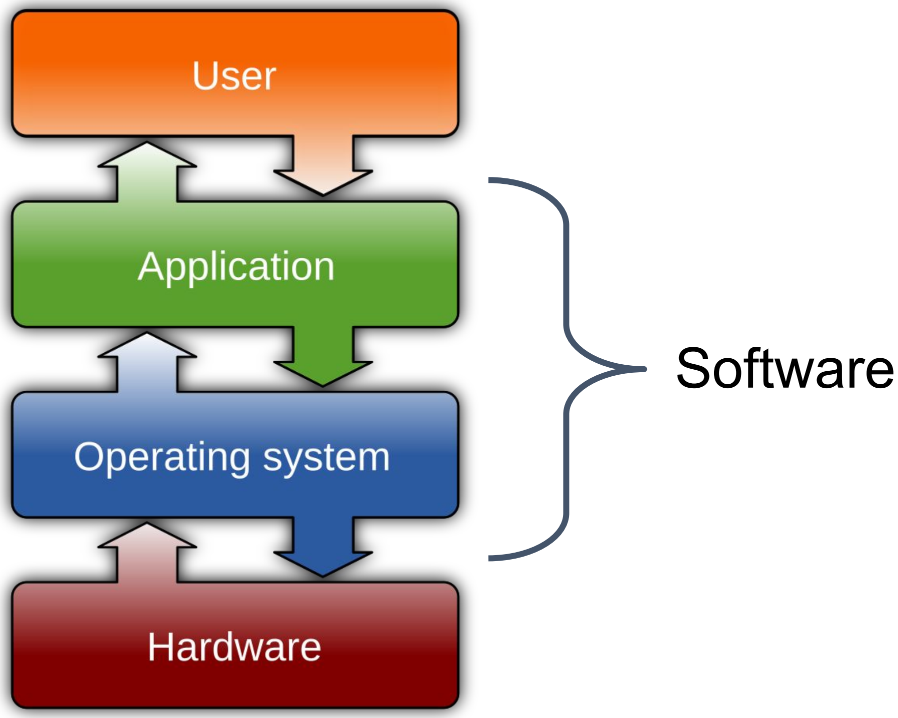


Source : Wikipedia (operating system)

Software

Set of programs written to perform special tasks or functions on a computer.

1. System Software : **Necessary software** to operate a device
2. Application Software : **Not necessary** to operate

- Text processors
- Web browsers
- Media players
- Games

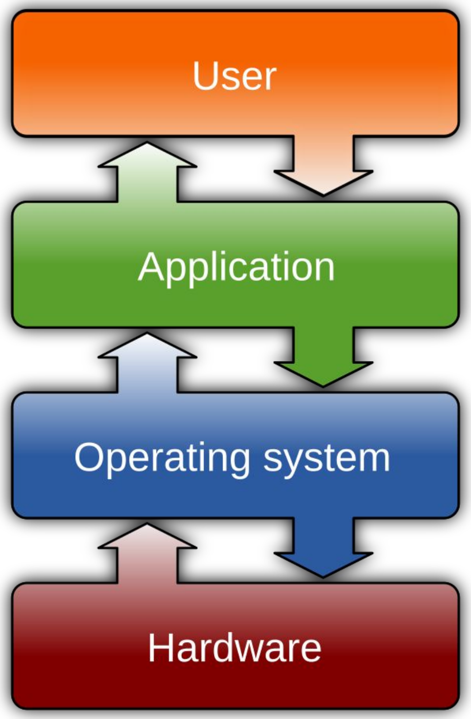
NB : WebApps vs. Desktop App

Operating Systems

An OS is a software used for interaction between the user and the computer hardware.

Give 4 examples?

System Software



Operating Systems

- Microsoft Windows
- MacOS
- Linux
- Android (let op!)

Types:

- Command Line
- GUI

Source : https://computerscience.chemeketa.edu/cs160Reader/OperatingSystems/OperatingSystems.html

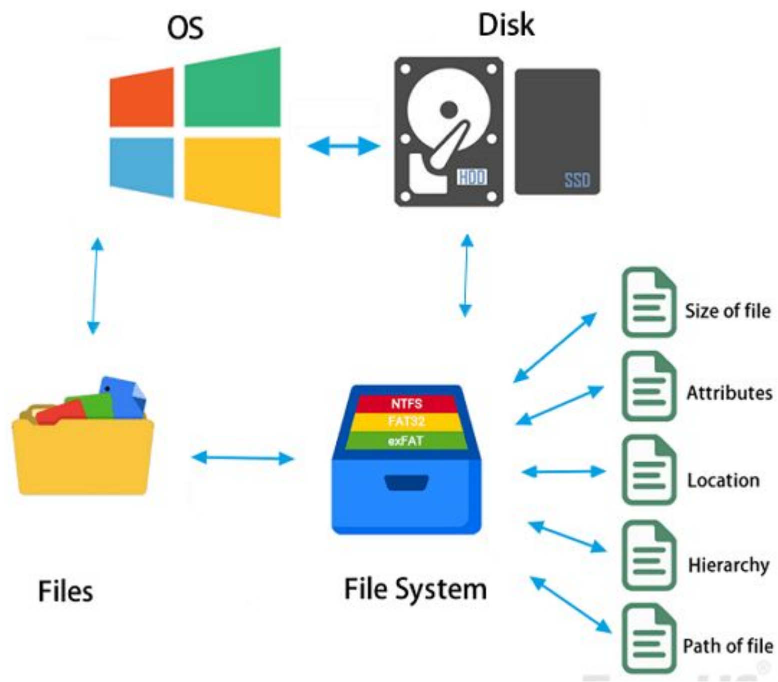
File systems

A **structure** used by an OS to organise and manage files on a storage device (HD, SSD, USB)

It defines how data is stored, accessed, and **organised on the storage** device.

Files, Directories, Metadata (name, size, metadata)

NB Your phone is also a file system!



Source : https://www.easeus.com/diskmanager/file-system.html

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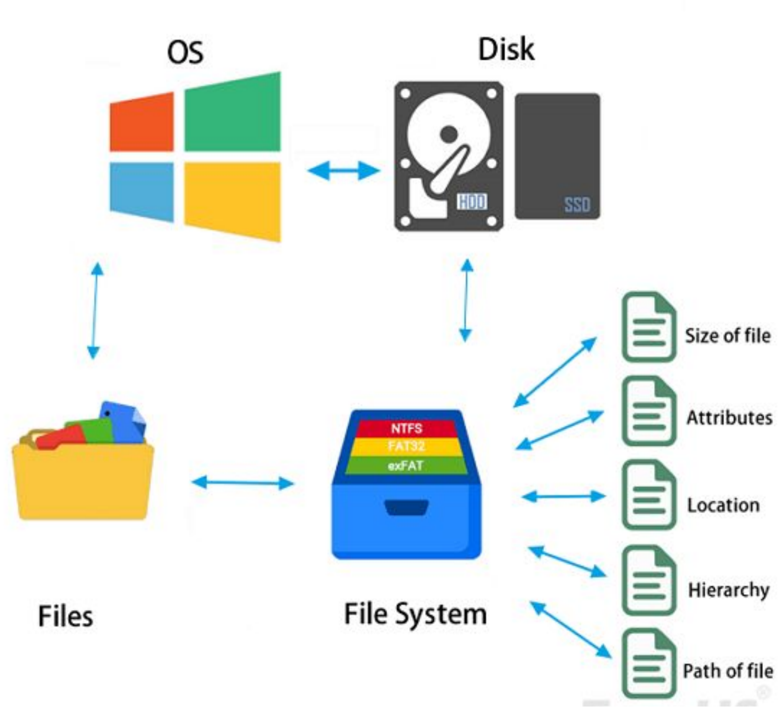

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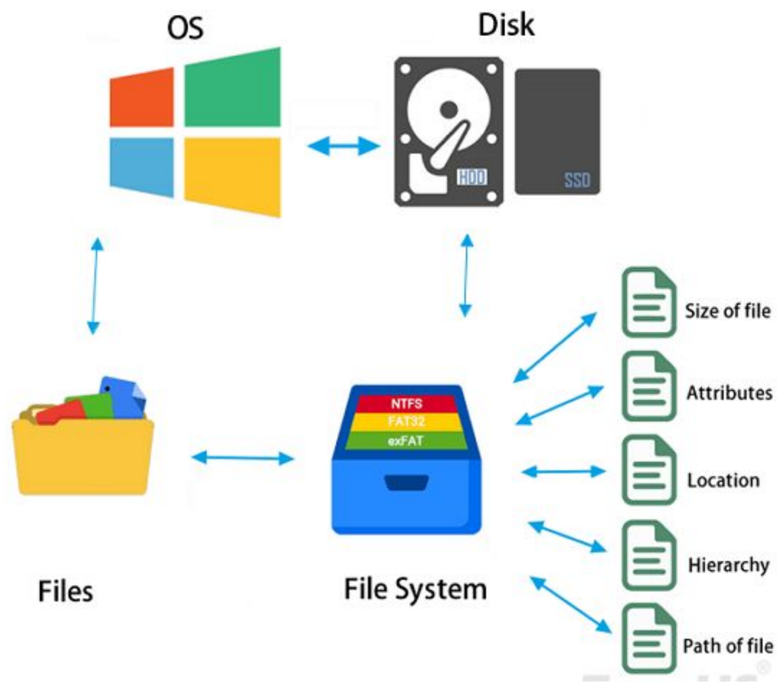


Source : https://www.easeus.com/diskmanager/file-system.html

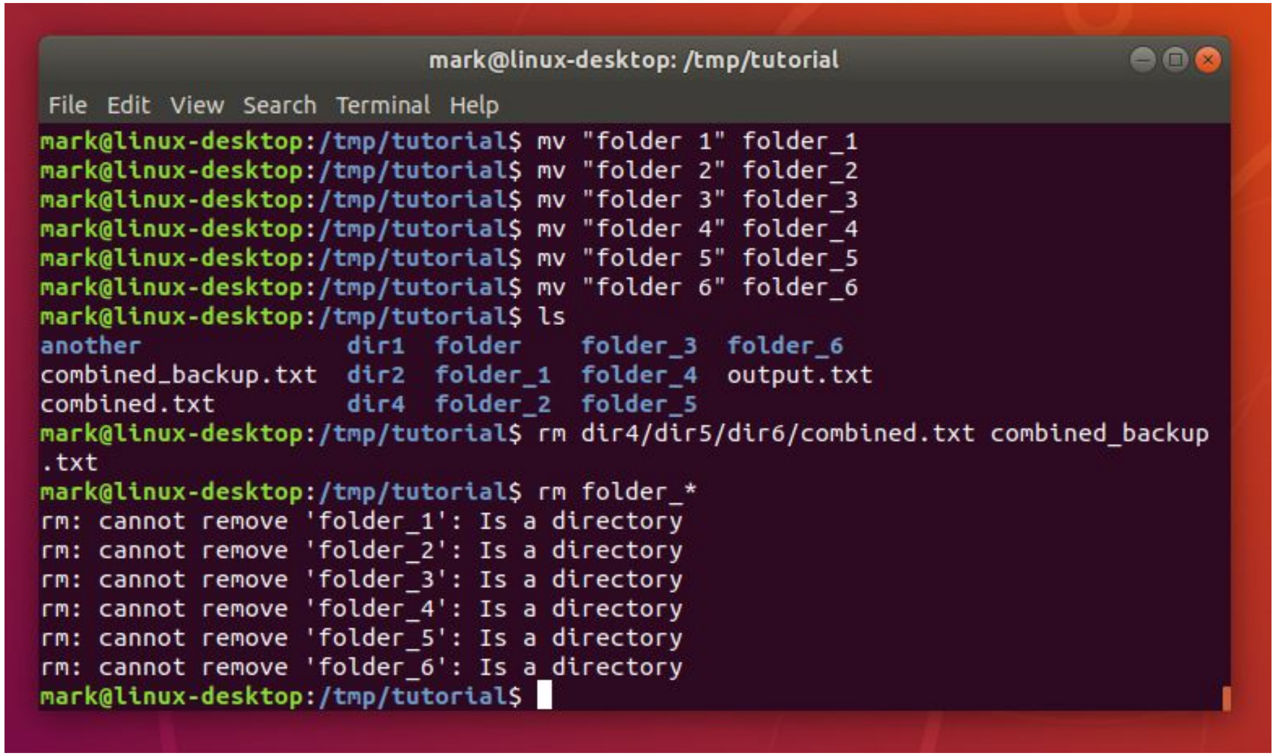
Working with files

You **need** to understand how to work with files and folders

- Folders : organise files
- HD : a tree of folders and files
- Applications : executable program that uses i/o files

Actions : **view, create, or edit** (cut, copy, paste,deleting) files.

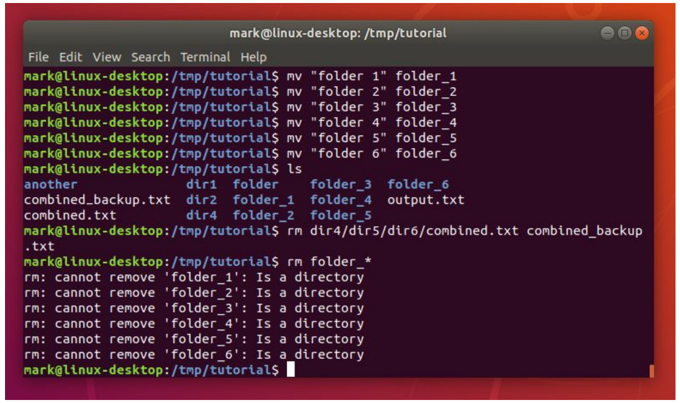
What’s this?



Source : https://ubuntu.com/tutorials/command-line-for-beginners

Terminal (= Command Line)

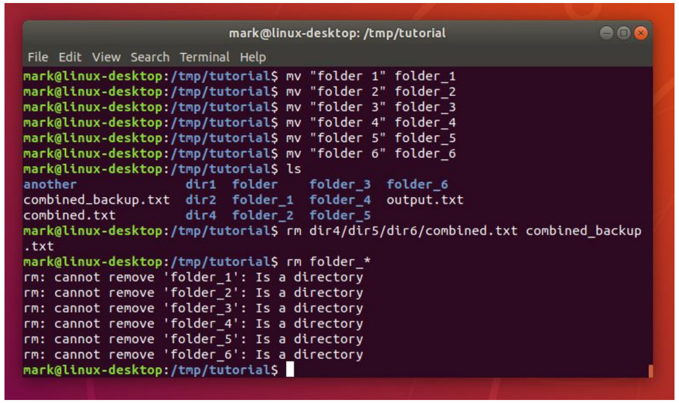
1. A **text-based method**/interface to access information on your computer
2. A text input and output environment
3. A program that acts as a **wrapper** and allows to enter commands that the computer processes



Terminal (= Command Line)

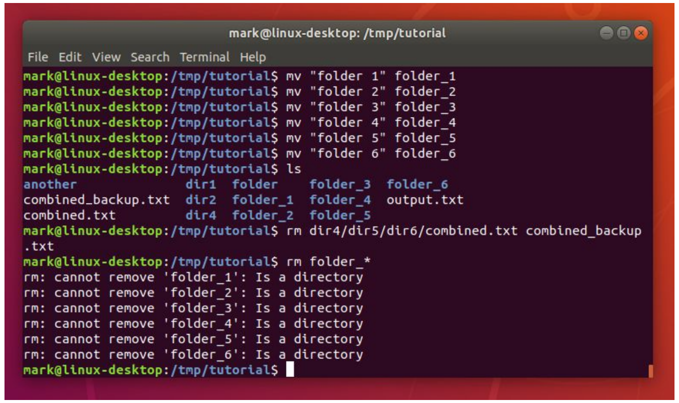
Actions:

- type commands
- manipulate files
- execute programs



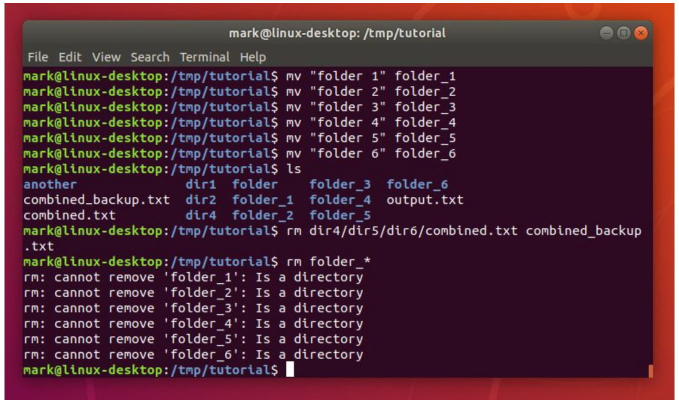
A terminal usually starts in the top-level directory of your user account!

Terminal : What for?



Terminal : What for?

- Faster, more efficient (i.e. creating sequences of commands to execute, i.e. a script)
- Task Automatisation (create a script today, run it tomorrow)
- Sometimes it is the only alternative (i.e. to communicate with remote servers)



Opening a Terminal

All operating systems come with a default terminal installed

Windows OS	Mac OS
write “cmd” in the search field or Type Windows+R keys	Press the "Command" key and the space bar
when the “run” appears, just type in “cmd”	type "Terminal" in the Spotlight bar
Hit return	Hit return

NB **different terminal languages**!! MacOs/Linux :Bash. Windows: Powershell.

Directories

You “move” around your file system content
current directory : your working directory

Directories are separated by a slash `/`

The topmost directory = a single backslash.

`/Volumes/ilaria/documents/Photos/`

Directories

Command + any modifiers/arguments required,
hit return to execute the command

Eg :

`cd /` = move to top directory

`cd ..` = go 1 directory level up

`cd /Volumes/ilaria/Documents/Photos/` = ?

Directories

`pwd` = print the complete pathname of the current directory

`ls` = listing documents

`ls .` = list all documents in current directory

`ls /Volumes/ilaria/Applications` = ?

Directories

`cd <your_directory>` = change directory (move into it).

no arguments (`cd`): go to top-level home dir

`cd .` = move to current directory (do not move)

`cd ..` = move to parent directory (go up one directory in the file tree)

Terminal Commands

`mv <from> <to>` : move a file from dir to dir

can also rename `mv myfile.txt myfile.csv`

`cp <from> <to>` : copy a file to a new location or name

`mkdir <directory name>` : make a directory

`rm` or `rmdir <directory name>` : remove a directory (the directory must be empty)

Files and strings

`touch <filename>` : create new file

`less <filename>` : scroll through a file

`cat <filename>` : send/show the file to standard output (into the terminal)

`echo "a string"` : print string in the standard output

Files and strings

```
Ilarias-Air:~ it2529$ echo "Hello"
Hello
Ilarias-Air:~ it2529$
```

Directing output: send stuff to a file using the greater than symbol
(1 time `>` : write a new file, 2 times `>>` : append to existing file)

```
echo "you are a smart student" > smart.txt
less smart.txt
```

Exercise

```
cd ..
cd <foldername>
mkdir newfolder
ping google.com
(ctrl + Z to stop )
```

Terminal Commands (Wildcards)

the star character `*` shows all the files in a directory

eg

List all files : `ls *`

List all files starting with A : `ls A*`

Never type `rm *`

Terminal Commands (Wildcards)

Tab completion

- when you have typed part of the name of a file, hitting the `tab` key will complete the filename as far as possible

eg. `/V<tab>i<tab>Doc>tab<` gets me `/Volumes/ilaria/Documents`

- If you have more files starting with the same sequence, tab completion will beep at you. If you hit tab again, then the terminal will show you the options

Terminal Commands (ShortCuts)

`Ctrl+ a`: move to the beginning of the line

`Ctrl+e` : move to the end of the line

`Ctrl+b` : move back a character

`Ctrl+f` : move fwd a character

Terminal Commands

Multiple commands together create a program (a small software)

A program can be saved as a **runnable script** file

Try

```
echo “Hello smart student” > my_script.sh
sh my_script.sh
```

Assignment

- Create a directory
- Create 1 or 2 text files (.txt)
- Paste text of your choice in the files
- Rename file
- Count words and list words by frequency
- Print a wordcloud with the list
- Save image
- Submit : the image (pdf, png) the frqList.txt file, the script (as a txt)

Coming Lectures

This Week

- Machine Learning by Peter Bloem
- Practice with Terminal
- Submit Literature Review!